Adam Rakhmanov

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Education

Duke University — Durham, NC
Bachelor of Science in Neuroscience and Computer Science — *Aug. 2026 – May 2030*

Ravenwood High School — Brentwood, TN
National Merit Finalist; ACT 35; GPA 4.72/4.00 — *May 2026*

Credentials

Biotechnology Aptitude and Competency Credential (BACC) — Issued Apr. 2026
Biotility, University of Florida — *Valid through Apr. 2031*

Research Experience

Student Researcher — Aug. 2023 – May 2026
Biomedical Sciences Research Program, Ravenwood High School — *Brentwood, TN*

- Designed and conducted two independent *Drosophila melanogaster* studies end to end, spanning hypothesis, experimental design, colony handling, data collection, analysis, and presentation
- Executed molecular workflows across a three-year sequence: DNA extraction, PCR, gene cloning, CRISPR editing, microarray gene-expression analysis, and dissection, under standard aseptic practice

Project Lead, NeuroSprout — 2025
Technology Student Association, Biotechnology Design — *1st Place State; Top 12 Nationally*

- Proposed a non-invasive approach to restoring CNS axon regeneration, targeting the oligodendrocyte-derived inhibitors Nogo-A, MAG, and OMgp that arrest the axonal cytoskeleton after injury
- Specified peptide-functionalized nanoparticles crossing the blood-brain barrier, dCas9 reversibly repressing inhibitor transcription, and split photoactivatable padCas9 gated by infrared light; rejected knockout, growth-factor, and stem-cell alternatives on glioma and immune risk

Projects

Quest Learning | *React, TypeScript, Supabase, Postgres, OpenAI API, Vercel* — Jul. 2024 – Present

- Engineered a platform translating cognitive-science principles such as spaced repetition and scaffolding into generated lessons, orchestrating multi-step LLM calls through serverless edge functions
- Designed the Postgres schema, authentication, and row-level security policies as co-founder and lead engineer; shipped 100+ resources and 110,000+ views into a 5,000-student pilot

FeelLoop | *Python, SQLite, OpenAI API, cron, multi-armed bandit* — 2025 – Present

- Built an autonomous content system that generates, schedules, and publishes across six accounts, closing the loop from creation through publication without manual steps
- Instrumented every post at the feature level and scored it on an engagement index tied to conversion, feeding a bandit allocator that steers each generation cycle across 772 posts and 277 metric snapshots

ExerciseBud | *React Native, Expo, Apple HealthKit, Google Health Connect, MongoDB* — 2024

- Integrated native step and activity streams from Apple HealthKit and Google Health Connect into a shared backend
- Generated individualized four-week training protocols with an LLM and built a real-time presence leaderboard

Experience

Software Engineer in Test — May 2022 – Jan. 2023
Novalab Tech — *Nashville, TN*

- Built automated test suites across front-end, back-end, and API layers for two enterprise products in Selenium (Java) and Cucumber
- Authored smoke and regression suites within an Agile team and maintained the manual test-case library

Honors

Breakthrough Junior Challenge Finalist: top 15 of 2,500+ international science-communication submissions (2025)
Technology Student Association: 6th Place Data Science and Analytics (National); 1st Place Biotechnology, Data Science, Geospatial Technologies (State)

Technical Skills

Languages: Python, Java, JavaScript, TypeScript, SQL (Postgres), HTML/CSS
Frameworks & Tools: React, React Native, Node.js, Supabase, MongoDB, SQLite, Selenium, Playwright, Git, Vercel
Laboratory: DNA extraction, PCR, gene cloning, CRISPR editing, gene-expression analysis, microarray, *Drosophila* handling
Research: Experimental design, primary-literature review, technical writing, scientific presentation, A/B and bandit testing
Spoken: English, Russian, Uzbek, Arabic (fluent in all four)